

Module -5

Incidence Matrix

Dr. Sathya P
Associate Professor
SENSE

Incidence Matrix

- Incidence matrix gives us the information about the branches, which are joined to the nodes and the orientation of the branch, which may be towards a node or away from it.
- Nodes of the graph form the rows and branches form the columns.
- If the branch is not connected to node, corresponding element in the matrix is given the value '**0**'.
- If a branch is joined, it has two possible orientations.
- If the orientation is away from the node, the corresponding matrix element is written as '**+1**'.
- If it is towards the node, the corresponding matrix element is written as '**-1**'.

- Incidence matrix is also known as augmented matrix.
- It is an $N \times B$ matrix with elements of $A_n = (a_{ij})$
 - $a_{ij} = 1$, when the branch b_j is incident to and oriented away from the i^{th} node.
 - $= -1$, when the branch b_j is incident to and oriented towards the i^{th} node.
 - $= 0$, when the branch b_j is not incident to the i^{th} node.
- Significance of the incidence matrix lies in the fact that it translates all the geometrical features in the graph into an algebraic expression.

- Using the incidence matrix, we can write KCL as

$$\mathbf{A}_n \mathbf{i}_B = 0, \text{ where } \mathbf{i}_B = \text{branch current vector.}$$

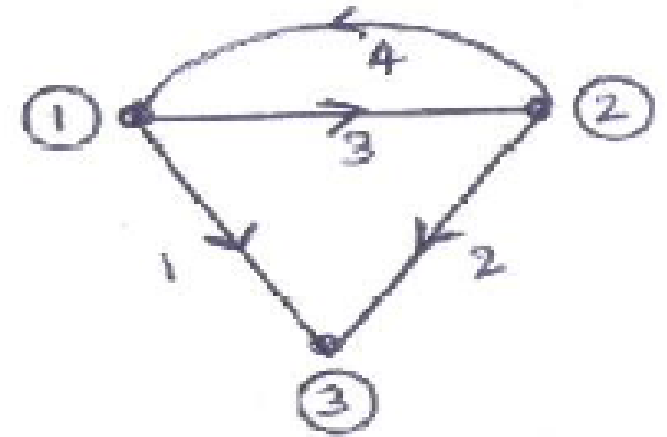
- But these equations are not linearly independent.
- The rank of the matrix $\mathbf{A}$ is $N - 1$.
- This property of $\mathbf{A}_n$ is used to define another matrix called reduced incidence matrix or bus incidence matrix.

Numerical - 1

Obtain the complete incidence matrix for the graph.

Solution:

Nodes	Branches			
	1	2	3	4
1	1	0	1	-1
2	0	1	-1	1
3	-1	-1	0	0



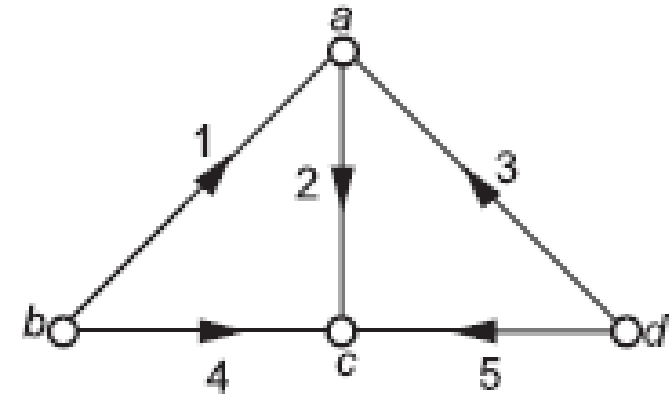
$$A_n = \begin{bmatrix} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 1 \\ -1 & -1 & 0 & 0 \end{bmatrix}$$

Numerical - 2

Obtain the complete incidence matrix for the graph.

Solution:

$$\mathbf{A}_n = \begin{array}{c} \text{Nodes} \downarrow \\ \begin{array}{c} a \\ b \\ c \\ d \end{array} \end{array} \begin{array}{c} \text{branches} \\ \begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \end{array} \end{array} \begin{bmatrix} -1 & 1 & -1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & -1 & -1 \\ 0 & 0 & 1 & 0 & 1 \end{bmatrix}$$



Note that sum of all elements in each column is zero

- **Properties of Incidence matrix:**

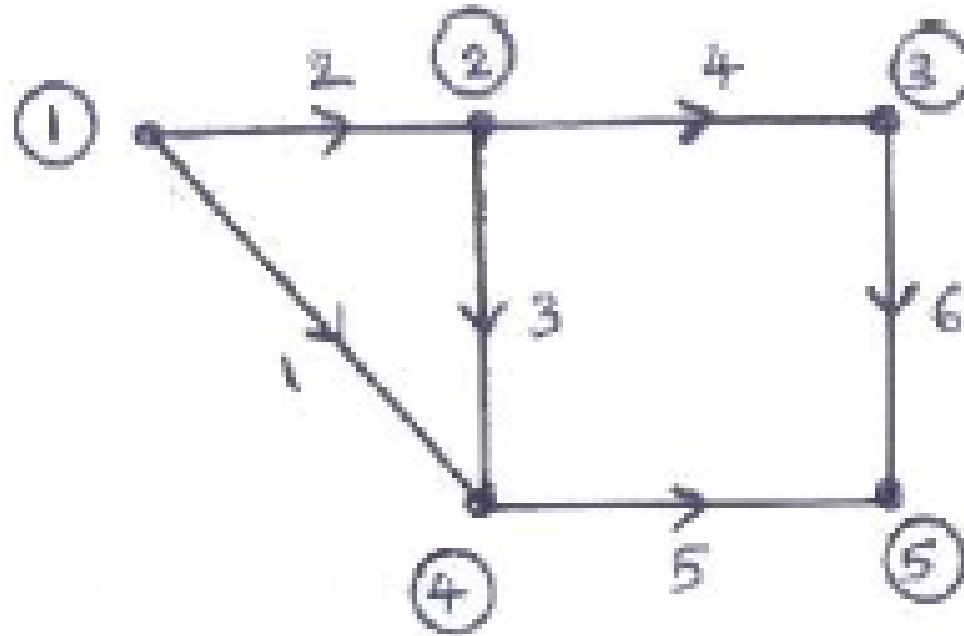
1. Each column has only two non-zero elements and all other elements are zero.
2. If all the columns of '**A_n**' are added, the sum will be a row whose elements equal zero.
3. If the graph has '**B**' branches and '**N**' nodes, the complete incidence matrix is of the order (N x B).

Reduced Incidence Matrix

- When one row is eliminated from the complete incidence matrix, the remaining matrix is called reduced incidence matrix.
- Any node of a connected graph can be selected as a reference node.
- Then the voltages of the other nodes (referred to as buses) can be measured with respect to the assigned reference.
- The matrix obtained from A_n by deleting the row corresponding to the reference node is the element Reduced incident matrix A and is called bus incidence matrix with dimension $(N-1) \times B$.
- A is rectangular and therefore singular.

Numerical - 3

Write the complete and reduced incidence matrix for the graph.



- Solution:

Nodes	Branches					
	1	2	3	4	5	6
1	1	1	0	0	0	0
2	0	-1	1	1	0	0
3	0	0	0	-1	0	1
4	-1	0	-1	0	1	0
5	0	0	0	0	-1	-1

Complete Incidence Matrix

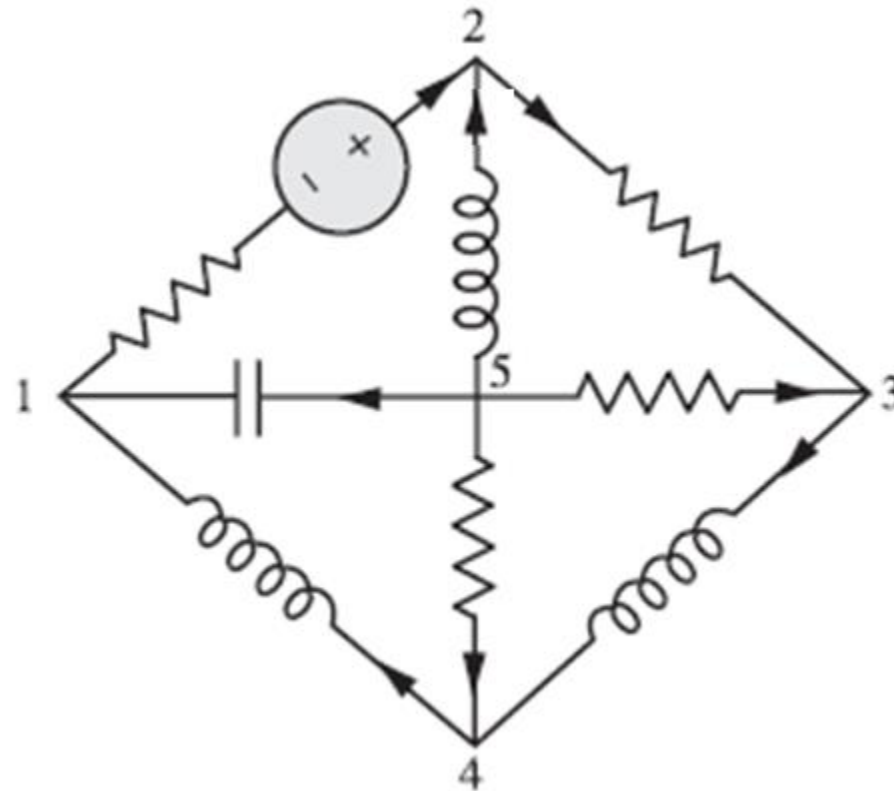
$$\mathbf{A}_n = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 1 \\ -1 & 0 & -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & -1 & -1 \end{bmatrix}$$

Reduced Incidence Matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 1 \\ -1 & 0 & -1 & 0 & 1 & 0 \end{bmatrix}$$

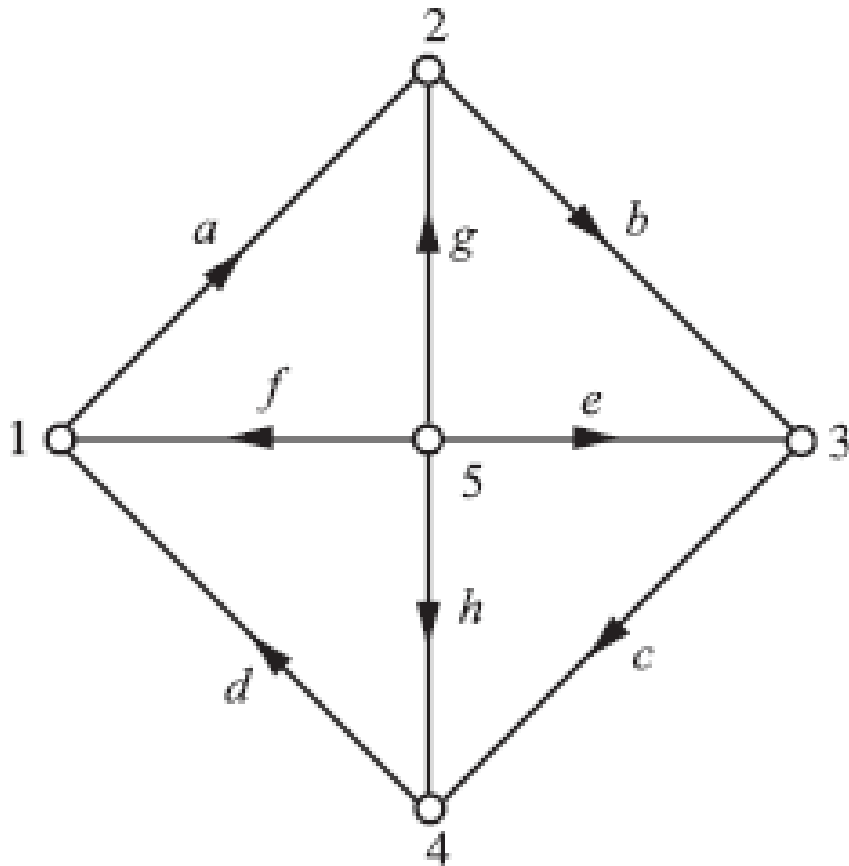
Numerical - 4

Write the complete incidence matrix for the graph.



- Solution:

Incidence matrix:



Nodes	Branch numbers							
	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>	<i>h</i>
1	+1	0	0	-1	0	-1	0	0
2	-1	+1	0	0	0	0	-1	0
3	0	-1	+1	0	-1	0	0	0
4	0	0	-1	+1	0	0	0	-1
5	0	0	0	0	+1	+1	+1	+1

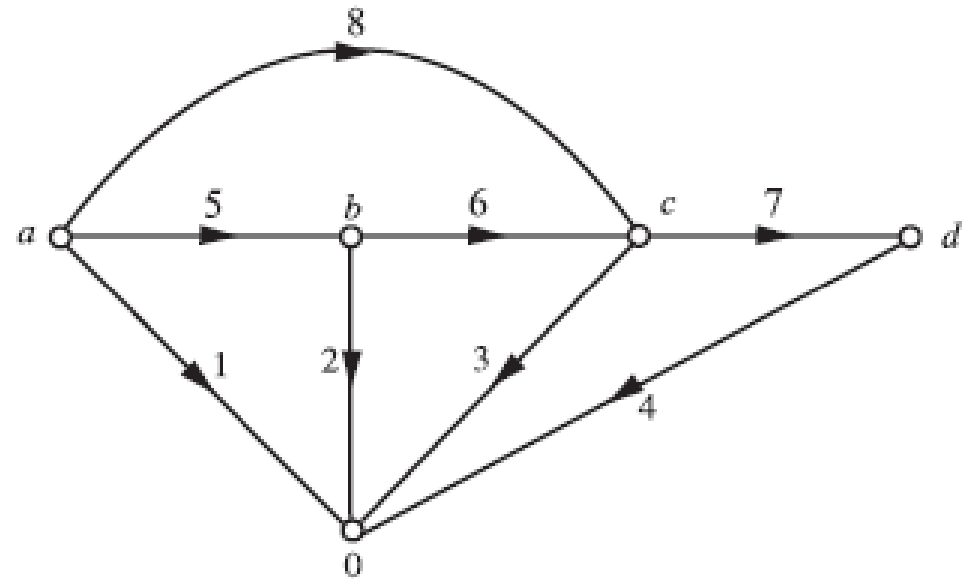
$$\mathbf{A}_5 = \begin{bmatrix} +1 & 0 & 0 & -1 & 0 & -1 & 0 & 0 \\ -1 & +1 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & -1 & +1 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & +1 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & +1 & +1 & +1 & +1 \end{bmatrix}$$

Numerical - 5

For the incidence matrix shown below, draw the graph.

$$\begin{array}{c} a \\ b \\ c \\ d \end{array} \begin{array}{cccccccc} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ \left[\begin{array}{cccccccc} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & -1 & 1 & -1 \\ 0 & 0 & 0 & 1 & 0 & 0 & -1 & 0 \end{array} \right] \end{array}$$

Solution:

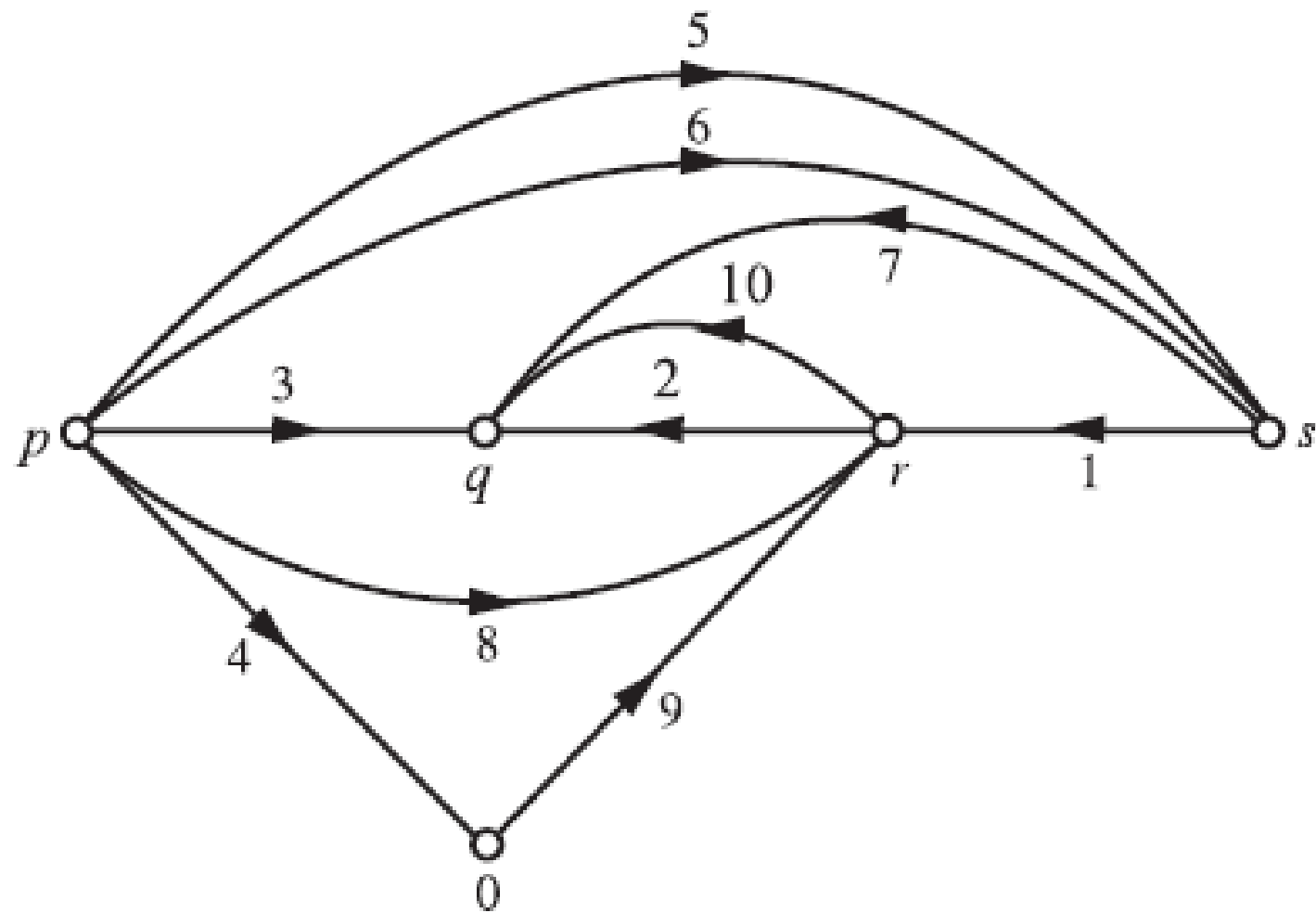


Numerical - 6

For the incidence matrix shown below, draw the graph.

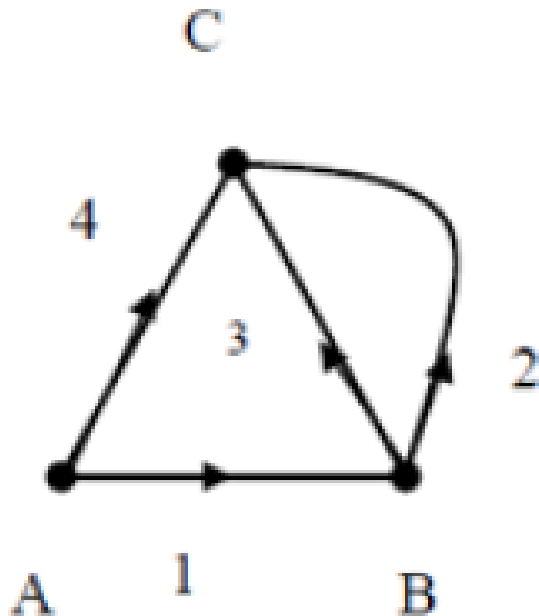
$$\begin{array}{c} p \\ q \\ r \\ s \end{array} \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & -1 & -1 & 0 & 0 & 0 & -1 & 0 & 0 & -1 \\ -1 & 1 & 0 & 0 & 0 & 0 & 0 & -1 & -1 & 1 \\ 1 & 0 & 0 & 0 & -1 & -1 & 1 & 0 & 0 & 0 \end{bmatrix}$$

Solution:



Property of Reduced Incidence Matrix

- The number of possible trees in a network is given by $A A^T$, where A is the reduced incidence matrix and A^T is the transpose of A .
- Numerical -7: Determine the number of trees in the given network:



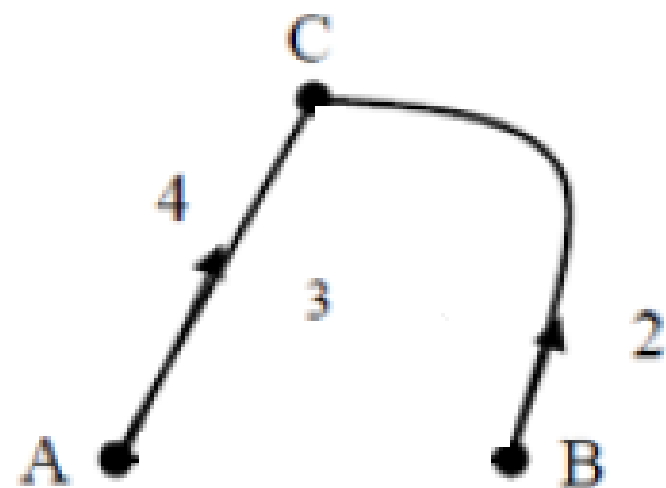
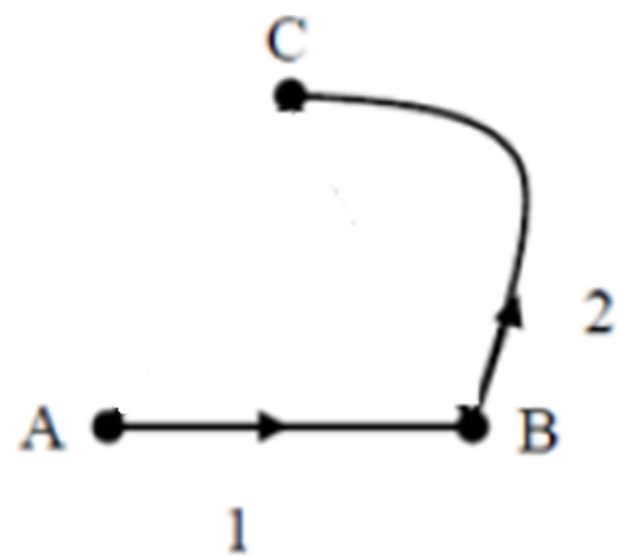
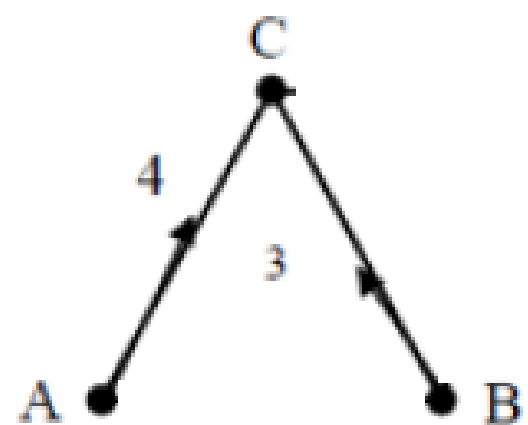
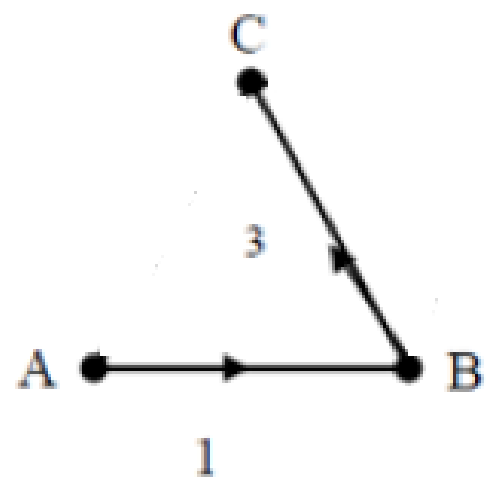
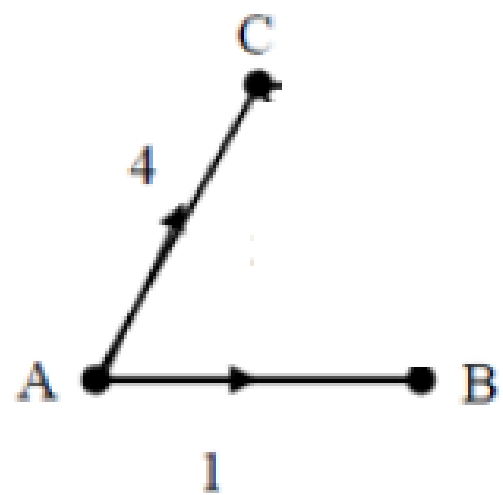
Node No.	Branches			
	1	2	3	4
A	1	0	0	1
B	-1	1	1	0
C	0	-1	-1	-1

$$A_3 = \begin{pmatrix} 1 & 0 & 0 & 1 \\ -1 & 1 & 1 & 0 \\ 0 & -1 & -1 & -1 \end{pmatrix} \quad [A] = \begin{pmatrix} 1 & 0 & 0 & 1 \\ -1 & 1 & 1 & 0 \end{pmatrix}$$

$$[A] [A]^t = \begin{pmatrix} 1 & 0 & 0 & 1 \\ -1 & 1 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 0 & 1 \\ 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1+1 & -1 \\ -1 & 1+1+1 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix}$$

$$| [A] [A]^t | = \begin{vmatrix} 2 & -1 \\ -1 & 3 \end{vmatrix} = (6-1) = 5$$

Number of possible trees are **5**.



Numerical -8

- For the given matrix, draw the oriented graph. How many trees are possible for this graph?

$$A = \begin{bmatrix} 0 & -1 & +1 & 0 & 0 \\ 0 & 0 & -1 & -1 & -1 \\ -1 & 0 & 0 & 0 & +1 \end{bmatrix}$$

- **Solution:**

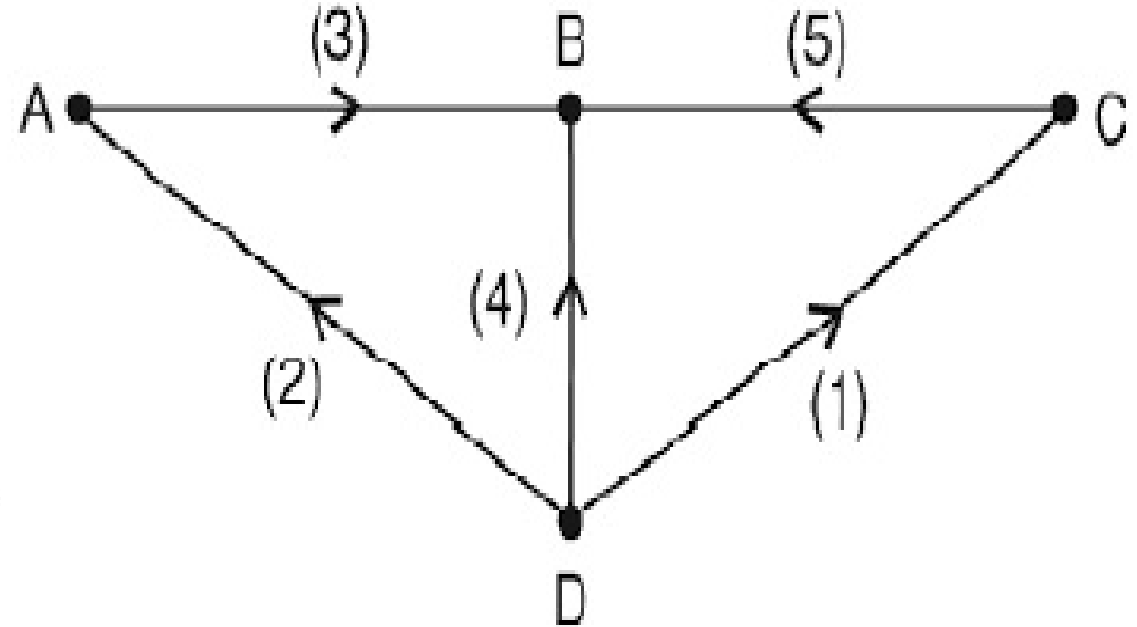
Step-I: To obtain complete incidence matrix.

For it adding one row in such way that sum of all entries in each column equals to zero.

$$\therefore A_c = \begin{array}{ccccc} & \text{Nodes} \downarrow & & & \\ \begin{array}{c} 0 \\ 0 \\ -1 \\ \hline +1 \end{array} & \begin{array}{c} -1 \\ 0 \\ 0 \\ +1 \end{array} & \begin{array}{c} 1 \\ -1 \\ 0 \\ 0 \end{array} & \begin{array}{c} 0 \\ -1 \\ 0 \\ +1 \end{array} & \begin{array}{c} 0 \\ -1 \\ +1 \\ 0 \end{array} \\ \begin{array}{c} A \\ B \\ C \\ D \end{array} & & & & \\ & \begin{array}{ccccc} (1) & (2) & (3) & (4) & (5) \end{array} & \leftarrow \text{Branches} \end{array}$$

Step-II: To obtain oriented graph.

- To mark nodes A , B , C and D arbitrary manner.
- Corresponding to first column: The non-zero (+ 1 and - 1) entries associated with nodes C and D . Therefore, branch (1) connected between C and D . Since, + 1 entry with node D , therefore direction of the branch will be from D to C , ($D \rightarrow C$). Similarly, the other branches are found out.



Step-III: Calculation of number of possible trees.

The no. of possible trees = $\det \{[A] [A]^T\}$.

$$\text{No. of trees} = \det \left\{ \begin{bmatrix} 0 & -1 & +1 & 0 & 0 \\ 0 & 0 & -1 & -1 & -1 \\ -1 & 0 & 0 & 0 & +1 \end{bmatrix} \begin{bmatrix} 0 & 0 & -1 \\ -1 & 0 & 0 \\ +1 & -1 & 0 \\ 0 & -1 & 0 \\ 0 & -1 & +1 \end{bmatrix} \right\}$$
$$= \det \begin{bmatrix} 2 & -1 & 0 \\ -1 & 3 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

$$= 2(6 - 1) + 1(-2 - 0) + 0(1 - 0)$$

$$= 10 - 2 = \mathbf{8}.$$